SARDet-CL: Self-Supervised Contrastive Learning With Feature Enhancement and Imaging Mechanism Constraints for SAR Target Detection

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***Abstract*— In recent years, supervised learning has seen notable progress in research on aircraft target detection using synthetic aperture radar (SAR) images, but its performance is heavily restricted by the amount of labeled data. Self-supervised learning (SSL) can address this problem by pretraining on unlabeled data to extract generalizable features. However, existing SSL methods are predominantly developed based on approaches designed for natural image processing, making them unsuitable for fully understanding SAR images. Therefore, this study proposes a self-supervised contrastive learning (SARDet-CL) method that integrates feature enhancement and imaging mechanism constraints, tailored to the characteristics and requirements of aircraft target detection tasks. Specifically, SARDet-CL enhances the model's ability to represent spatiotemporal features through spatial feature masking and multitemporal consistency learning, and robustly extracts target features using a dynamic threshold quantization strategy. In addition, structural information is inferred based on the SAR imaging mechanism and incorporated as an auxiliary self-supervisory signal to guide the model toward a more accurate understanding of SAR images. Experimental results based on various SAR datasets show that the proposed method outperforms other pretraining techniques in downstream SAR target detection tasks.**

***Index Terms*— Aircraft target detection, objective function, pretext task, self-supervised learning (SSL), synthetic aperture radar (SAR).**

I. Introduction

SAR systems are widely utilized in remote sensing surveys due to their ability to provide continuous observation, unaffected by weather conditions. Aircraft targets, being high-value objects, have emerged as a focal point in synthetic aperture radar (SAR) image interpretation research. Recently, supervised learning has had significant accomplishments in detection and classification by extracting high-level features from large labeled datasets. Subsequently, numerous deep learning-based methods have been developed for SAR aircraft target detection. However, supervised learning relies heavily on high-quality labeled data. When trained on limited labeled samples, the model is prone to overfitting due to the narrow scope of the dataset, which leads to performance degradation. Although the expansion of labeled samples can alleviate this problem to a certain extent, constructing high-quality and large-scale labeled datasets consumes great human cost because the labeling of SAR image targets requires professional interpretation knowledge.

SSL offers a novel approach in natural image processing by first extracting meaningful representations through self-supervised signals from the image data and then fine-tuning on limited labeled samples to enhance downstream task performance.

This study focuses on contrastive learning (CL), a well-established self-supervised technique in natural image processing [1]. Fig. 1 illustrates the basic working principle of CL. The core principle involves constructing pairs of positive and negative samples through pretext tasks. The objective function provides an incentive signal to the model, encouraging it to extract features that can differentiate positive from negative samples. This approach aims to learn features that are both invariant and distinctive.

Recent studies have successfully applied CL to optical remote sensing, demonstrating its effectiveness in tasks such

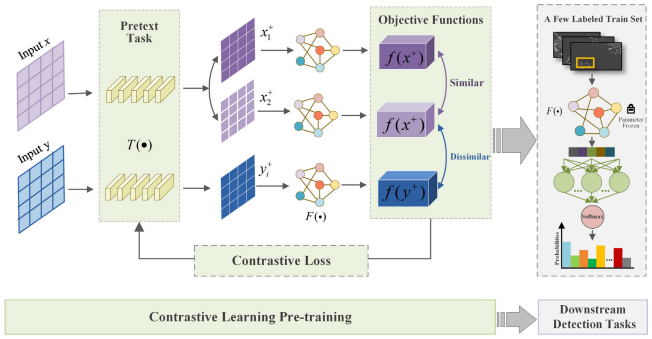


Fig. 1. Workflow: positive and negative samples are constructed through pretext tasks, and their positive and negative sample information is used as the self-supervisory signal of the objective function so that the model can learn distinguishable features.

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The pretrained model weight of SARDet-CL can be accessed at https://github.com/yangyifremad/SARDet-CL

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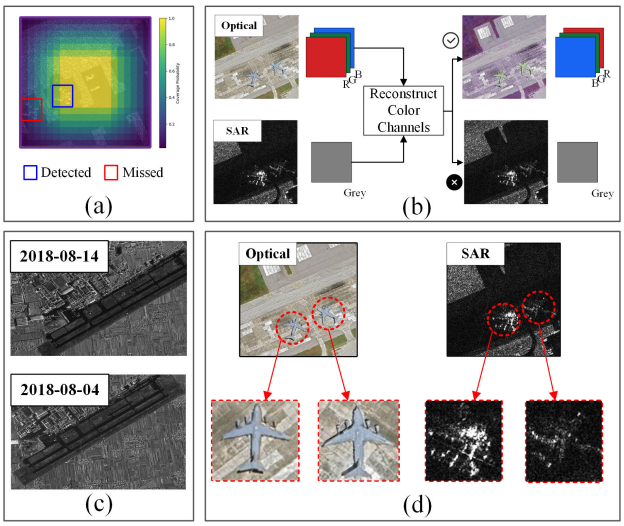


Fig. 2. Challenges of SSL in SAR image aircraft target detection. (a) Aircraft are sparsely distributed. (b) Failure of aircraft feature enhancement. (c) Interference of multitemporal information. (d) Large differences in features of similar aircraft targets.

as target classification and semantic segmentation. Since both natural and optical remote sensing images adhere to similar imaging mechanisms, employing CL on optical remote sensing datasets has proved highly effective.

However, the application of CL to SAR image target detection remains largely unexplored. As illustrated in Fig. 2, this study explores the key challenges of CL in SAR object detection.

A. Center Inductive Bias

The targets of SAR images appear randomly in various regions of the image. The detection task requires accurate extraction and analysis of local features and location box regression. Current CL methods are mainly designed based on the ImageNet dataset and its classification tasks, where the target objects are predominantly found in the image's central region. The self-supervisory signals in these methods generally emphasize the image's primary features [2], that is, the center inductive bias, as shown in Fig. 2(a). Ignoring local regions, especially image edge features, can lead to missed detection of targets.

B. Failure of Target Feature Enhancement

The pretext task in CL imposes effective perturbations on images through feature enhancement strategies to construct semantically consistent but visually different positive sample pairs, thereby guiding the model to learn generalizable feature representations. In optical image processing, the rich color information provided by the three RGB channels offers natural redundancy for augmentation operations. Techniques such as channel weighting and color jitter have been proved to be among the most effective self-supervised augmentation methods [3]. However, this enhancement strategy becomes ineffective for SAR images. In single-channel SAR imagery, RGB color space operations cannot be applied, making it impossible to construct meaningful positive sample pairs, as shown in Fig. 2(b). This limitation reduces the effectiveness of the self-supervised learning (SSL) signal.

C. Interference of Multitemporal Information

As illustrated in Fig. 2(c), SAR data often includes temporal information, indicating that for a given location $(\text{lat}_i, \text{lon}_i)$, there exists a sequence of spatially aligned images over time $X_i = (x_{t1}, \ldots, x_{ti})$, that is, multitemporal samples. Such samples from the same location usually contain highly similar semantic features. However, in the design of the self-supervised signal for CL, distinct images are considered negative samples relative to each other. In this scenario, spatially aligned multitemporal samples can mistakenly be used as negative samples, disrupting the learning process. This incorrect feedback interferes with, or even undermines, the ability of the model to effectively extract features from the data.

D. Feature Differences Caused by Imaging Mechanism

SAR images currently use active detection, backscattering, and coherent imaging as their main technical forms. This mechanism will lead to significant differences in the projection of targets and their shadows. As shown in Fig. 2(d), visually identical or similar physical structures can appear vastly different when imaged. Therefore, it is not applicable to rely solely on the 2-D information directly obtained from the image to guide the SSL process in SAR images.

In the face of these challenges, based on the requirements of aircraft target detection tasks, SARDet-CL was proposed. To make the pretext task compatible with downstream detection tasks, the adversarial mask module and dynamic quantization (Dync-Quantization) module are designed. The adversarial mask module focuses on reducing missed target detection. Dync-Quantization flexibly adopts sample information with close spatial distribution to enhance the target feature. As the most direct part of the self-supervisory signal, the objective function introduces effective spatiotemporal information and structural information. The former is generated by the joint action of geographic location information and the SAR-SIFT algorithm [4], and the latter is inverted by the neural radiation field model that follows the SAR imaging mechanism [5].

The main contributions of this work are as follows.

1) A method for SSL from unlabeled SAR images, SARDet-CL, is proposed. This method first performs pretraining on unlabeled data. It then significantly improves downstream target detection performance under limited labeled conditions, thus getting rid of the dependence on large-scale labeled data.
2) A new pretext task is designed to guide the model to focus on a broader range of global features by applying a mask to the spatial distribution, thereby reducing central inductive bias. Meanwhile, a Dync-Quantization enhancement strategy based on geographic information adaptively optimizes target features and improves the model's generalization ability for unstable SAR targets,

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enabling the learning of more robust and reliable SAR target features.

3) A new objective function is proposed, where spatially aligned images from different time points are treated as positive samples to enhance the model's feature consistency across varying imaging conditions. Additionally, based on the SAR imaging mechanism, the structural similarity (SSIM) loss is extracted as a constraint signal to guide the model in learning feature representations that better align with physical principles.

The remainder of the article is organized as follows. Section II reviews the related research works on SAR aircraft target detection and SSL. Section III introduces the proposed SARDet-CL method in detail. Section IV verifies the superiority of SARDet-CL through experiments, and Section V concludes.

II. Related Works

A. Research on SAR Aircraft Target Detection Methods

Target detection is a key technique in the decoding task of SAR images. Since 2014, with deep learning's remarkable successes in computer vision, it has become the dominant approach in the field of SAR image interpretation [6].

Jia et al. [7] developed an aircraft component characterization model, utilizing multiscale features to capture the overall outline and key components of the aircraft. Suo et al. [8] improved aircraft detection performance by proposing an adaptive image enhancement algorithm that suppresses speckle noise and enhances scattering features. Sun et al. [9] proposed an SAR target generation method that integrates deep imaging geometric information to improve target recognition accuracy. Xiang et al. [10] constructed a feature image representing the superposition of sidelobes, which achieved better suppression of strong cross-shaped sidelobes and significantly improved the accuracy of target detection. Additionally, numerous studies on SAR target detection have employed deep learning methods like the YOLO series, Fast-RCNN, oriented R-CNN (Ori-RCNN), and Swin Transformer.

However, these current target detection methods mainly belong to the supervised learning domain, and their training process relies heavily on the guidance of manually labeled information.

B. Research on SSL

SSL can extract valuable information from many unlabeled samples, reducing the need for labeled data. In the realm of natural image processing, several successful SSL methods have emerged, such as simple CL (SimCLR) [3], momentum contrast (MoCo v2) [11], swapping multiple views contrastive learning (SWAV) [12], simSiamese representation learning (SimSiam) [13], and bootstrap your own latent (BYOL) [14].

Recent studies have extended SSL to optical remote sensing, demonstrating its feasibility. Liu et al. [15] employed principal component analysis on multispectral images to create two enhanced views and maximized consistency between them during training for SSL. Tao et al. [16] utilized SSL to interpret optical remote sensing images and established a visual model with high generalization capability. These methodologies are typically suited to optical remote sensing tasks, where data characteristics are analogous to those of natural images.

However, such learning paradigms are not universally applicable, especially when the data characteristics significantly differ from optical data. Currently, the effective application of SSL in SAR images is somewhat limited, focusing primarily on registration, segmentation, and classification tasks. Ren et al. [17] introduced multi-image factor SSL to facilitate directional feature learning and obtain generalized features, significantly enhancing pixel-level terrain classification with minimal patch-level labeled data. Li et al. [18] proposed a high-performance joint self-supervised architecture for SAR target recognition tasks, overcoming the interference of speckle noise. Wang et al. [19] built an efficient pretraining model based on cross-domain continuous pretraining of multispectral and SAR.

To date, the application of SSL to object detection tasks in SAR images remains largely unexplored.

III. Methodology

In this section, the proposed SARDet-CL method is elaborated upon. The overall framework of SARDet-CL is depicted in Fig. 3. Initially, the application of CL in computer vision tasks is discussed in Section III-A. Subsequently, the two main components of SARDet-CL are introduced in Sections III-B and III-C: the pretext tasks and the objective functions. Finally, the overall network architecture and the corresponding algorithm are described.

A. Contrastive Learning

The core concept of self-supervised CL is that features derived from similar data should exhibit similarity, with feature vectors being as close as possible in the feature space. Conversely, feature vectors derived from dissimilar data should be as distant as possible, thereby enabling the learning of both invariant and distinguishable feature expressions [1]. As shown in Fig. 1, the implementation process primarily involves two steps: the pretext task and the objective function.

1) Pretext Task: It defines positive and negative samples. For each unlabeled data point x, data augmentation techniques are employed to construct a positive sample pair (x, x^+), where $x^+ = T(x)$ is derived by applying a random data transformation $T(x)$ to x (such as cropping, scaling, rotation, random noise, and color distortion). All other samples in the dataset (y, y^+) are treated as negative samples.

2) Objective Function: It defines the goal of the model learning task as positive sample matching and negative sample staying away. The CL model is guided to continuously optimize the parameters during the training process by calculating the feature similarity loss.

B. Pretext Tasks in SARDet-CL

Different SSL signal designs originate from unique starting points, leading to varied focuses on the features learned. For instance, in the CL of target recognition tasks, random cropping serves as an effective strategy for view

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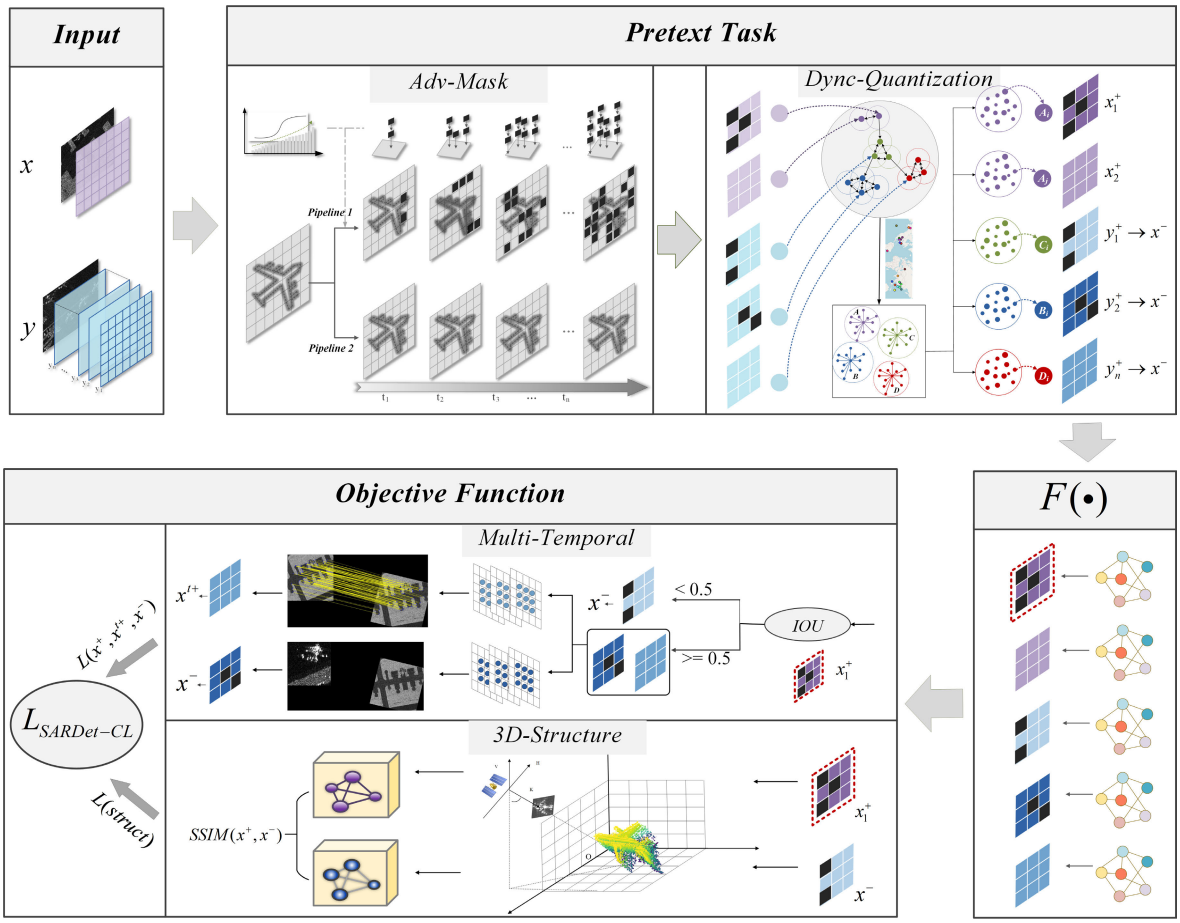


Fig. 3. Workflow of SARDet-CL. The pretext task consists of two modules: adv-mask and Dync-Quantization. Objective function consists of two modules: Multitemporal and 3D-Structure.

enhancement, reinforcing the focus on central image features. However, this approach is not well-suited for target detection tasks.

In this section, the pretext tasks appropriate for SAR aircraft target detection are explored, considering both the types of tasks and the unique characteristics of SAR data.

1) Mask Learning Strategy Based on Adversarial Training: In this comparative learning work, a masker-encoder architecture is designed to address the problem of central imputation bias introduced by traditional random cropping. The masker applies a Bernoulli mask to randomly and selectively obscure certain patches of the image, where the masking probability is governed by a Bernoulli distribution. This approach determines which patches are retained and which are obscured. The encoder then processes only the visible patches, transforming the input image into a representation in a high-dimensional feature space. This selective visibility is critical as it forms the foundation of the entire mask enhancement process.

Specifically, as shown in Fig. 3, pipeline1 adds a masker M, the feature of the masked pipeline is represented as Z_m, pipeline2 does not cover any patch, and the feature is represented as Z_v. Our goal is to learn how to make Z_m consistent with Z_v through the encoder F. That is,

$$F(Z_m) = F(Z_v). \tag{1}$$

In the encoding representation space, the transition from visible patches to masked patches compels the model to perform semantic inferences, which are crucial for feature comparison. This approach designs an encoder that not only needs to learn the global representation of the complete image but also to be able to extract valid feature information even when some patches are masked. This improves the model's ability to capture features from all regions of the image, especially in the noncentral region. As illustrated in Fig. 4, utilizing the masking strategy allows for the learning of global information across the entire image, in contrast to traditional random cropping.

The masker functions as a structural perturbation to the input data, distorting the image by masking certain patches and thereby complicating the encoder's task of feature extraction to differentiate between positive and negative samples. The greater the number of patches masked by the masker, the more challenging the task for the encoder becomes, establishing an adversarial relationship between the masker and the encoder. We introduce an adversarial strategy for mask learning that progressively increases the complexity of the learning task. This incremental challenge enhances the model's robustness and adaptability in complex environments.

In practice, a masking function $M_t(x)$ is applied to each input x. The function $M_t(x)$ is defined in

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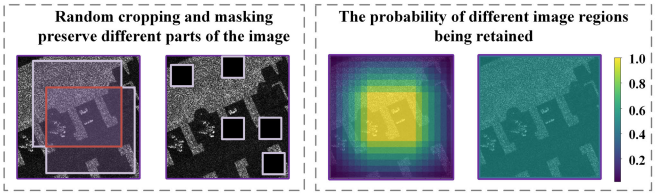


Fig. 4. Difference between random cropping and masking. (a) Random crop. The central area is easily included. (b) Mask. Randomly adopt each patch of the image. (c) Random crop. Unable to focus on edge areas. (d) Mask. Able to focus on the overall situation.

the following equation:

$$M_t(x) = x \odot (1 - m_t) \tag{2}$$

where $\odot$ denotes element-wise multiplication and m_t is a mask generated randomly according to a Bernoulli distribution with parameter p_t. The parameter p_t changes dynamically over time and is defined by the formula

$$p_t = \frac{1}{1 + e^{-k(t-T/2)}}. \tag{3}$$

In this equation, k is a scaling factor and T is the total number of training epochs. This formulation ensures that as training progresses, the probability of masking increases gradually, forcing the model to focus more on unmasked features. This approach not only enhances the robustness of the model by exposing it to different parts of the data but also mitigates the issue of overfitting by preventing the model from relying too heavily on specific features, such as patches that are frequently unmasked.

2) SAR Dync-Quantization Module Guided by Spatial Information: Due to the scattering characteristics of aircraft components and their material properties, aircraft targets usually exhibit a higher backscatter coefficient than the background. Specifically, flat surfaces such as wings produce surface scattering; the vertical junction between the aircraft and the ground causes double-reflection scattering; and the multilayered complex structure of the engine forms volume scattering. These scattering characteristics provide a key basis for target detection and recognition.

In this process, scattering-induced phase delays lead to unstable echo distributions. Coherent superposition causes random amplitude variations, diminishing target–background contrast. Moreover, phase inconsistencies and angular variations exacerbate feature instability across imaging conditions.

This study introduces dynamic threshold quantization into the pretext task to enhance the scattering characteristics of the target and improve the model's adaptability to the coherent superposition effect, as shown in Fig. 5(a). Considering that spatially close areas usually have similar landforms and overall scattering characteristics, DBSCAN [20] clustering is performed on samples based on geographic information, and thresholds are dynamically obtained from samples in the same cluster (i.e., samples with similar spatial distribution). The detailed process is as follows:

$$p_i' = Q(p_i, \mu) \tag{4}$$

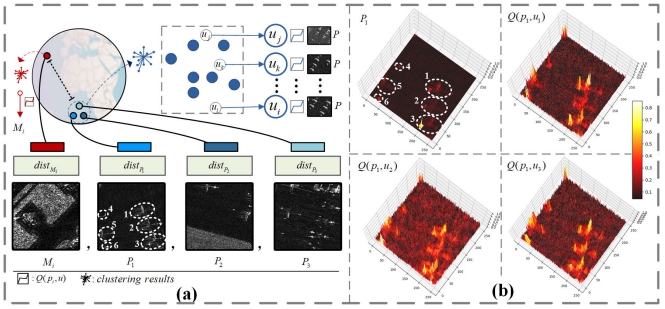


Fig. 5. (a) Dynamic quantification based on geographic information clustering: P_1, P_2, and P_3 show that the same type of targets can exhibit varying backscattering intensities in different areas. (b) After Dync-Quantization, the target features are enhanced to varying degrees.

$$Q(p_i, \mu) = \begin{cases} G_{\max}, & \text{if } p_i > \mu \\ \dfrac{p_i - P_{\min}}{P_{\max} - P_{\min}} \cdot \Delta + G_{\min}, & \text{otherwise} \end{cases} \tag{5}$$

$$\Delta = \frac{G_{\max} - G_{\min}}{P_{\max} - P_{\min}} \tag{6}$$

$$\mu = \text{random}(u_{\text{DBSCAN}}) \tag{7}$$

$$u_{\text{DBSCAN}} = \{q \in D \mid \text{dist}([\text{lat}_i, \text{lon}_i], [\text{lat}_j, \text{lon}_j]) < \varepsilon\}. \tag{8}$$

In the context, p_i represents the pixel value in the original image, while p_i' denotes the quantized pixel value. μ is a threshold determined based on DBSCAN clustering. The values $P_{\min}$ and $P_{\max}$ represent the minimum and maximum across the entire image. $G_{\min}$ and $G_{\max}$, respectively, are the minimum (usually 0) and maximum (usually 255) target grayscale levels. Δ is the quantization interval, ensuring that pixel values are linearly mapped from the original data range to the target grayscale range. μ is randomly sampled from a set of candidate thresholds based on geographic information, utilizing DBSCAN to cluster SAR images based on geographic location information (latitude and longitude). DBSCAN can effectively handle noisy datasets and identify clusters of arbitrary shapes. In (8), D represents the set containing all data points, dist is the distance function between two points, and ε is the clustering neighborhood radius. The DBSCAN algorithm works by identifying areas of high-density connectivity and finding core points within those areas. Starting from these core points, the algorithm includes all directly density-reachable points within the ε-neighborhood into the same cluster. This approach eliminates the need to predefine the number of clusters, making it especially suitable for applications in SSL.

The aforementioned quantization steps are incorporated into the pretext task of CL, as illustrated in Fig. 5. Specifically, for the same sample p_i, information from samples of the same category is dynamically adapted for preprocessing in its two data enhancement pipelines. Consequently, this adjustment allows for the modification of (5) to

$$p_{i_1}' = Q(p_i, \mu = \mu_{p_1}) \tag{9}$$

$$p_{i_2}' = Q(p_i, \mu = \mu_{p_2}). \tag{10}$$

In this process, μ_{p1} represents the threshold selected by the first pipeline, while μ_{p2} denotes the threshold chosen by

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the second pipeline. The SSL signal encourages the model to learn similar features in p'_{i_1} and p'_{i_2}. This approach implies that during the learning process, the model not only identifies the unique features of each sample but also discerns the common features of neighboring samples as indicated by clustering. Such adaptation to positive samples with varying scattering intensities significantly enhances the model's robustness and generalizability.

As shown in Fig. 5(b), P_1 is an unquantized image, and the distinction between the aircraft target and the background is minimal. After three different threshold quantizations, the target features are enhanced to varying degrees. Among them, under the parameter setting of u_3, target 2 is effectively enhanced, but under the parameter setting of u_1 calculated by P_1 itself, the enhancement effect of target 2 is limited. By dynamically setting the threshold, the model can adapt to samples with different scattering intensities, improving the robustness and generalization of the model.

C. Objective Function in SARDet-CL

To address the issue of interference from multitemporal information and large differences in target features, the objective function is redesigned in SARDet-CL, which mainly includes two aspects: effective multitemporal (Multitemporal) information learning and feedback of 3-D structural (3-D-Structure) information. The details are as follows.

1) Multitemporal Information Learning: This involves mining spatially aligned multitemporal samples, correcting them to positive samples, and encouraging the model to implicitly learn additional spatiotemporal information. First, the candidate samples are initially screened. The four vertices latitude and longitude information of the image is utilized to calculate the IOU. IOU is obtained from (11). When the IOU exceeds 0.5, the sample is considered a viable multitemporal candidate

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}}. \tag{11}$$

Second, considering that in the process of acquiring SAR images, due to the influence of factors such as slight changes in satellite orbits, differences in sensor angles, and atmospheric conditions, the images may have different degrees of spatial offset. When the offset is small, the process can be considered to be approximately in line with the idea of data enhancement in CL and is an effective alignment sample. Still, when the spatial offset is large, such as when there is no overlapping area, it cannot be considered a valid temporal positive sample pair. The accuracy of longitude and latitude information cannot meet the alignment requirements of SAR data. Existing studies have proposed various advanced registration techniques tailored to the characteristics of SAR images. In this study, however, multitemporal positive samples only require the presence of overlapping regions without the need for pixel-level accuracy. Consequently, a relatively straightforward SAR-SIFT algorithm is employed to extract and match key points.

The SAR-SIFT algorithm incorporates a novel gradient calculation method, specifically designed for SAR images. This approach enhances the robustness of gradient direction and magnitude in the presence of multiple scattering and strong noise, making it more effective at handling multiplicative noise caused by scattering. Additionally, it reduces errors stemming from the scattering characteristics of electromagnetic waves. The working method of SAR-SIF is shown in the following equation:

$$C_{\text{SH}}(x, y, \alpha) = \mathcal{G}_{\sqrt{2}\alpha} * \begin{bmatrix} (G_{x,\alpha})^2 & (G_{x,\alpha} \cdot G_{y,\alpha}) \\ (G_{x,\alpha} \cdot G_{y,\alpha}) & (G_{y,\alpha})^2 \end{bmatrix} \tag{12}$$

$$R_{\text{SH}}(x, y, \alpha) = \det(C_{\text{SH}}(x, y, \alpha)) - d \cdot \text{tr}(C_{\text{SH}}(x, y, \alpha))^2. \tag{13}$$

In the formula, $C_{\text{SH}}(x, y, \alpha)$ represents the structure tensor at the position (x, y) and scale α, capturing the local gradient information of the image. $\mathcal{G}_{(2)^{1/2}\alpha}$ is a Gaussian kernel with a standard deviation of $(2)^{1/2}\alpha$, mainly used for smoothing the image. $G_{x,\alpha}$ and $G_{y,\alpha}$ represent the gradients of the image at scale α in the x- and y-directions, respectively, calculated from the Gaussian-smoothed image. $R_{\text{SH}}(x, y, \alpha)$ is the response function, which detects features such as corners by combining the determinant and trace of the structure tensor. $\det(C_{\text{SH}}(x, y, \alpha))$ denotes the determinant of the structure tensor, measuring the intensity of local variations in different directions, while $\text{tr}(C_{\text{SH}}(x, y, \alpha))$ is the trace of the structure tensor, representing the overall variation in intensity in the local region. The parameter d is a constant that balances the contributions of the determinant and trace in the response function, ensuring sensitivity to both edges and corners.

Successfully matched candidate samples were fed back as positive samples, and in this way, the TimeInfoNCE loss was designed to encourage the model to implicitly learn additional spatiotemporal information.

Given a data sample x, the loss calculation process of the TimeInfo objective function is expressed as the following equation:

$$S^+ = \exp\big(s(x, x^+)/\tau\big) + \exp\big(s(x, x_t^+)/\tau\big) \tag{14}$$

$$L_{\text{TimeInfo}} = -\log \frac{S^+}{S^+ + \sum_{i=1}^{N} \exp(s(x, x_i^-)/\tau)}. \tag{15}$$

In the context of training, x^+ represents the positive samples constructed through pretext tasks, while x_t^+ denotes the temporally aligned multitemporal positive samples. N indicates the number of negative samples and x_i^- stands for the negative samples. The cosine similarity is represented by s and τ is the temperature hyperparameter. The goal is to minimize the loss function, aiming to bring the positive samples closer together and push the negative samples further apart.

We argue that utilizing real multitemporal positive images can encourage the network to learn better semantic representations of real data, as opposed to relying solely on positive samples constructed from pretext tasks.

2) 3-D Structural Information Compensation: In remote sensing images, many scenes are similar but show significant differences after imaging, so it is often difficult to rely solely on 2-D visual information to guide SSL in SAR images. To this end, this study characterizes structural information based on the mapping and projection (MPA) imaging mechanism [5] and incorporates SSIM loss into the objective

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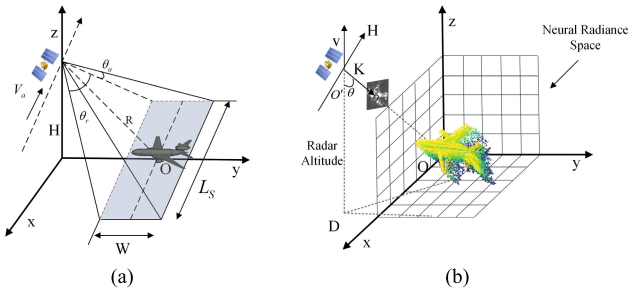


Fig. 6. Based on NERF, voxel rendering is used to represent 3-D information. (a) SAR imaging geometric model. (b) Radar 3-D voxel coordinate system.

function, aiming to use physical and geometric properties to enhance the model's understanding and generalization ability of the complex and variable characteristics of SAR target features. The process consists of two parts.

a) Construction of SAR-NERF: In our previous work [21], based on the radiation transfer theory and the MPA algorithm in the imaging simulation process, the SAR neural radiated field (SAR-NERF) was established to characterize the structural characteristics of SAR images. First, the SAR imaging space is divided into voxel space, and the characterization process is shown in Fig. 6.

Voxel Space: SAR utilizes pulse compression and synthetic aperture techniques to achieve 2-D high-resolution imaging, as illustrated in Fig. 6(a). The geometric relationships in the side-view strip diagram are influenced by factors such as orbital altitude, incident angle, and azimuth. NERF is a ray tracing-based 3-D voxel modeling method. As shown in Fig. 6(b), during the imaging simulation, it is assumed that each unit grid of the imaging space corresponds to a voxel, and the contribution of each row in every azimuth resolution interval is computed individually. Then, the scattering coefficient of a single voxel can be expressed as $I_{i,j,q}$.

MPA Imaging Process: After obtaining the scattering contribution of a single voxel, the voxel space is rendered and imaged. In the MPA imaging mechanism, the scattering imaging in the incident plane is achieved through mapping projection. The radar differentiates targets based on their distance, such that targets located at the same distance are mapped to the same pixel in the resulting image. The scattering amount in the same pixel is accumulated to obtain the total scattering coefficient of the pixel. The scattering strength of the pixel unit at (i, j) can be expressed as

$$I_{i,j} = \sum_{q=0}^{N_\theta - 1} I_{i,j,q}. \tag{16}$$

Learning of Structural Information: After rendering the intensity image, the neural radiation field can automatically fit the 3-D voxel information by minimizing the error between the predicted intensity and the actual value to achieve the representation of geometric structure information.

b) Introducing structural loss into the objective function: After obtaining the SAR-NERF representation of the 3-D structure, based on (15), we introduce the feedback of structural information and add a term related to SSIM when calculating the similarity of negative samples. In this way, the images originally defined as negative samples may have different penalty strengths according to their SSIM. Specifically, the formula can be modified as follows:

$$\text{struct}(x, x_i^-) = \text{SSIM}([I_x], [I_{x_i^-}]) \tag{17}$$

$$L_{\text{SARDet-CL}} = -\log \frac{S^+}{S^+ + \sum_{i=1}^{N} \left[\exp((s(x, x_i^-) + \lambda \cdot \text{struct}(x, x_i^-))/\tau\right]}. \tag{18}$$

Among them, struct represents the SSIM of the sample, which is measured by the SSIM index [22]. $[I_x]$ and $[I_{x_i^-}]$ are the backscattering coefficients of the voxel space scattering energy mapped and projected into pixels for samples x and x_i^-, respectively. λ is a hyperparameter used to control the influence of SSIM on loss. When the SSIM between negative samples and positive samples is high, the influence of SSIM reduces the penalty for being classified as a negative sample. This adjustment allows CL to treat negative samples not strictly as "negative," but to provide an additional feedback signal based on SSIM. This feedback, which contains physical and geometric information, can significantly improve the model's understanding and generalization capabilities regarding the complex and variable features of SAR images.

D. Proposed SARDet-CL

In summary, according to the requirements of the target detection task, SAR scattering law, and imaging mechanism, SARDet-CL is designed. Algorithm 1 gives the detailed steps of the proposed SSL framework.

IV. Experiments and Analysis

This section first describes the details related to the experimental setup such as dataset and evaluation metrics. Then, extensive comparative experiments are conducted on the proposed method. Second, the impact of data domain differences on method performance is further explored. Third, the performance of each method in real large scene images was tested. Finally, the effectiveness of each module of SARDet-CL is ablated and investigated.

A. Experimental Data

This study aims to fully utilize the embedded information of the SAR dataset to improve the capability of aircraft target detection. For this purpose, a dataset was constructed based on GF-3 SAR satellite data that can be used for self-supervised training. The training process is conducted without labeling information. This dataset is also manually labeled for performance testing of the target detection task. In addition, validation of the target detection task was performed on four other publicly available and widely used SAR aircraft detection datasets to test the performance of the pretrained model migrated to a nonhomologous dataset. It is worth mentioning that all four publicly available datasets have been quantized and the images do not contain geographic information. An overview of the four datasets is provided below.

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Algorithm 1 SARDet-CL Framework

Input: SAR image dataset D, mask rate p_t, model parameters Θ, quantization threshold μ
Output: Trained model $F(\Theta)$
Begin
  For image $X \in D$ **Do**
    $[\mu_1, \mu_2] = Random(\mu_{DBSCAN})$
    Pipeline 1:
      $X_m = Mask(X, p_t)$
      $X^+ \leftarrow X_1^+ = Quantize(X_m, \mu_1)$
      $update(p_t) \uparrow$
    Pipeline 2:
      $X^+ \leftarrow X_2^+ = Quantize(X, \mu_2)$
  End For
  For other images $Y \in D$ **Do**
    $X^- \leftarrow Y$
    If $Alignment(Geo(X), Geo(y)) == True$ **Then**
      $X^+ \leftarrow X_t^+ = Y$
    Else
      $X_{3D} = NeRF(X)$
      $Y_{3D} = NeRF(Y)$
      $L_{struct} = \lambda \cdot SSIM(X_{3D}, Y_{3D})$
    End If
  End For
  $L^+ = Similarity(F(X_m^+; \Theta))$
  $L^- = Similarity(F(X_m^-; \Theta)) + L_{struct}$
  $L = L^+ + L^-$
  $F(\Theta) \leftarrow optimizer.update(L, \Theta)$
  Return Trained model $F(\Theta)$
End

1) GF-3 SSL Dataset: The dataset consists of 120 spotlight-mode full-scene images from GF-3 satellites, with HH and VV polarizations. These images predominantly cover aircraft target parking areas at major airports such as Beijing Capital Airport and Shantou Waisha Airport. The dataset comprises 2491 images, each 256 × 256 pixels, encompassing 5305 aircraft targets across 16 different categories. Each image retains its original single-channel 16-bit amplitude information, along with geographic information encoding and time stamps of the captures.

2) SAR-AIRcraft-1.0 Dataset [23]: The dataset is based on GF-3 satellite spotlight single-polarization images. Released in 2023, the dataset comprises 4368 images, which include seven aircraft categories and a total of 16 463 aircraft targets. It is characterized by a large coverage area, complex backgrounds, dense target distributions, and multiple scales.

3) MSAR-1.0 Dataset [24]: This dataset is based on a combination of GF-3 and HISEA-1 satellite images, released in 2022. HISEA-1 is a C-band VV polarimetric SAR satellite. The dataset encompasses 6368 aircraft.

4) SADD Dataset [25]: The dataset is constructed from images of the TerraSAR-X satellite, released in 2022. It contains 2966 images featuring 7835 aircraft targets with distinct structures and outlines, varying in size and set against complex backgrounds, including airport runways and aprons.

5) ZKXT Dataset [6]: The dataset consists of 2000 images in seven categories. These images cover a variety of challenging and complex scenes, such as building sidelobe noise interference, aircraft feature discreteness, and so on.

B. Evaluation Metrics

Common evaluation metrics for object detection include AUC, average precision (AP), recall, and $F1$-score [26], [27]. In this study, we focus on instance-level detection and adopt two widely used metrics—AP at IoU threshold 0.5 and $F1$-score—to evaluate the performance of pretrained models on downstream detection tasks. The expression is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{21}$$

$$\text{AP} = \int_0^1 P(R)\, dR. \tag{22}$$

The terms TP, FP, TN, and FN stand for true positive, false positive, true negative, and false negative, respectively.

C. Implementation Details

The work is based on the Ubuntu 16.04 system and carried out on 2 NVIDIA RTX 3090 GPUs using the PyTorch framework.

In the SSL stage, the SARDet-CL performs pretraining based on the encoder F(·) of resnet50, which is consistent with the backbone used by typical CL methods such as SimCLR and MoCo-v2 [11], [12], [13], [14]. The masker parameters were optimized, and finally, an 8*8 grid was used to divide the image into 64 patches. The probability of each patch being masked is determined by (3). In the adversarial training process, the upper bound of the mask rate is set to 0.7. When employing the DBSCAN clustering algorithm to analyze image geographic information, a neighborhood radius of 10 km is established. The training configuration includes setting the number of epochs to 400 and a batch size of 256. Additionally, the LARS optimizer is employed, starting with a learning rate of 0.0001 and a weight decay coefficient of 0.0000015. This setup helps to regularize the model and prevent overfitting. Additionally, due to the initial instability of the CL model during training, a cosine annealing learning rate scheduler is applied throughout the 400 epochs to manage this concern. Finally, the SARDet-CL framework is trained with a temperature τ of 0.2 and a λ of 0.5. This pretraining process is based on dataset 1.

In the fine-tuning stage of the object detection task, two representative target detection networks, Fast-RCNN [28] and Ori-RCNN [29], were used for the downstream tasks. The former is a horizontal box detector, while the latter specializes in rotation box detection. By default, only the encoder part from the pretraining stage is loaded to initialize the backbone of the detector. The AdamW optimizer is utilized for all fine-tuning processes, which occur over 36 epochs using a batch

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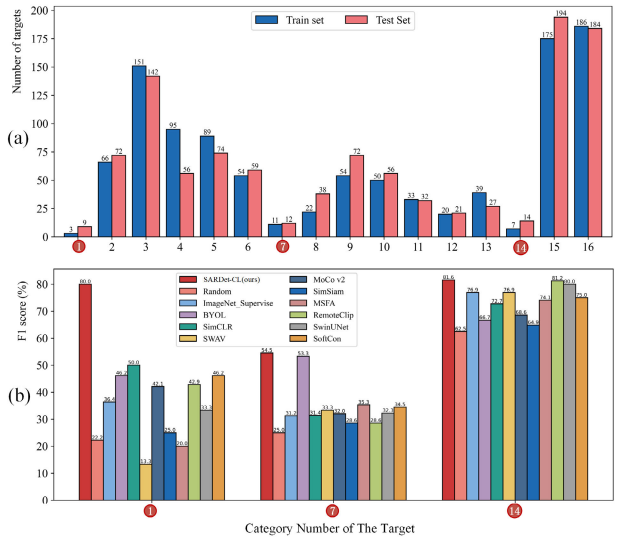


Fig. 7. (a) Training set utilizes only 20% of the labeled data from the complete dataset, with categories 1, 7, and 14 having particularly few training samples. (b) Performance of these few-shot categories under the Ori-RCNN detector.

size of 16. The learning rate is initially configured at 0.0001, incorporating a linear warm-up period at the beginning of training. Subsequently, a step-by-step learning rate scheduling strategy is applied, reducing the learning rate to one-tenth at the 24th and 33rd epochs.

D. Experimental Results

Based on the experimental settings described, the proposed SARDet-CL method was pretrained on the information-rich GF-3 SSL dataset. During this process, no target annotation information is required; the model relies solely on the data itself. The performance of SARDet-CL will be fully validated from the following aspects.

1) Comparison With Other Methods: During the fine-tuning training phase of the object detector, only 20% of the labeled data is used for training, and of the remaining data, 20% of the total dataset is randomly selected for performance testing. The distribution of each category in the training and test sets is illustrated in Fig. 7(a). Notably, the sample size for most targets in the training set is even smaller than that in the test set. This includes categories 1, 7, and 14, which are particularly scarce and represent a significant few-sample problem. Performance tests were conducted based on two typical target detectors: the Mask-RCNN and the Ori-RCNN.

In the performance comparison experiments, a network with randomly initialized parameters serves as the basic baseline. That is, the parameters are randomly initialized without using any pretrained model. Additionally, the results were compared with those from a pretrained model on the ImageNet dataset, which is widely used in computer vision research. Furthermore, several other advanced pretraining methods such as SimCLR [3], MoCo v2 [11], SWAV [12], SimSiam [13], BYOL [14], RemoteClip [30], SwinUnet [31], SoftCon [19], and MSFA [32] are also used for comparison. The first five methods have been verified as effective methods in a large number of studies. RemoteClip is a novel remote sensing multimodal training model, and the last three methods are the latest pretraining models applied to SAR images.

TABLE I
COMPARISON OF DIFFERENT METHODS ON ORI-RCNN AND FAST-RCNN (20% TRAINING DATA FINE-TUNING)

Method	Ori-RCNN		Fast-RCNN	
	AP (%)	F1 (%)	AP (%)	F1 (%)
Random-Baseline	38.36	40.29	35.20	36.20
ImageNet-Baseline	60.10	57.80	55.19	55.28
SimCLR	61.45	58.14	54.67	58.98
MoCo v2	53.23	50.78	53.30	54.30
SWAV	55.72	54.48	50.23	54.95
SimSiam	41.97	41.83	40.13	41.66
BYOL	63.55	60.19	58.06	57.59
RemoteClip	56.21	53.16	51.92	50.75
SwinUNet	53.46	51.99	54.96	52.98
SoftCon	60.81	58.70	51.09	51.30
MSFA	64.06	58.71	58.91	58.50
SARDet-CL (Ours)	**70.29**	**67.17**	**64.21**	**63.57**

Table I presents the experimental results. The results after introducing the pretrained model are all better than the strategy of randomly initializing the parameters, which verifies that the pretraining process is important for the fine-tuning of the target detector. Moreover, the SARDet-CL method significantly surpasses the ImageNet pretraining approach. Although the ImageNet pretrained model is trained using millions of data and their annotation information, far exceeding the self-supervised SAR data used in our experiment, the distinct characteristics between optical and SAR images limit its effectiveness in improving SAR image detection. SSL methods, on the other hand, can efficiently learn the unique features of SAR data at a lower cost. Additionally, the performance of different SSL methods varies widely, and inappropriate SSL methods can produce ineffective learning. For instance, MoCo v2, SWAV, and SimSiam performed worse than the ImageNet pretrained model when applied to SAR data. Overall, SARDet-CL achieved the best results on both detectors, with a performance improvement of approximately 6% over the second-best method.

To specifically observe the impact of the pretrained model on the performance of the detection network during fine-tuning, Fig. 8 illustrates the performance of each stage of the detector training process on the test set. By analyzing the indicator, SARDet-CL achieves leading performance early in the training process, demonstrating that the features learned by our pretrained model are more effective for target detection tasks. Additionally, Fig. 7(b) presents the $F1$ scores of three few-shot target categories under different strategies. SARDet-CL consistently outperforms others, particularly on target type 1, which has only three training samples, achieving more than a 30% improvement over other methods. Some visualization results are also shown in Fig. 9 for qualitative analysis.

In Fig. 9(a), due to the characteristics of SAR imaging, the features of the same type of targets are different. The SARDet-CL method shows better generalization, while other methods have missed detection and false alarms. In Fig. 9(b),

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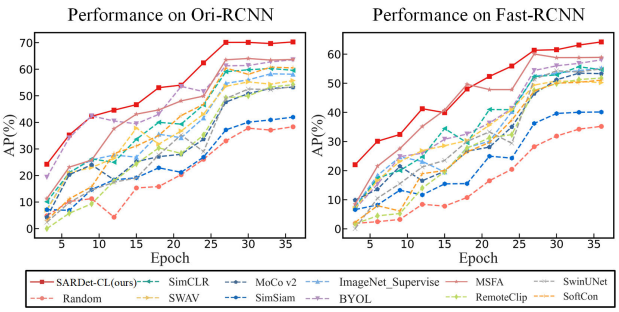


Fig. 8. Details of the fine-tuning process of different models, showing the changes in AP performance at each epoch during the fine-tuning process.

the target features are greatly disturbed by the sidelobe noise generated by the building, but SARDet-CL can still achieve stable performance. We believe that this is because the introduction of multitemporal information enables the model to extract similar features in the aligned images of different times with and without noise. In Fig. 9(c), it was found that the backscattering intensity of the two identical targets is significantly different. While some algorithms can detect brighter targets, only the SARDet-CL method successfully identifies the relatively darker target, owing to the Dync-Quantization strategy introduced in the pretext task. In Fig. 9(d), only SARDet-CL can find the target at the edge of the image. As shown in Fig. 10, feature heat maps were visualized using the Grad-CAM method to analyze the model's attention to features at different locations. It was observed that SARDet-CL's attention mechanism can better focus on edge targets and effectively distinguish targets from the background. Other methods tend to focus on main features or larger buildings, overlooking edge aircraft targets and the surrounding background, which leads to difficulties in detection. Overall, SARDet-CL achieves the best performance.

In addition, when there is sufficient training data, that is, all labeled data except the test set is used for fine-tuning, the pretrained model should also improve the performance of the detector. Experimental verification has also been conducted. The results are shown in Table II. After adding labeled data for fine-tuning, the performance gap of the detector is narrowed due to the addition of a large amount of supervised information. Even the randomly initialized baseline method has achieved a high performance. However, the SARDet-CL method maintains its advantage, as the pretext task and objective function designed for this approach enable it to extract more relevant information from unlabeled data, which is effective for the object detection task.

It is worth mentioning that the images employed for self-supervised CL pretraining in this section closely resemble those used in downstream tasks and are sourced from the same dataset.

2) Performance on Nonhomologous Datasets: This section explores the impact of domain differences between the self-supervised dataset and the downstream task fine-tuning dataset on model performance. The performance of pretrained models migrated to nonhomologous datasets is tested on four publicly available and widely used SAR datasets. Table III qualitatively analyzes the similarity of the datasets in terms of the SAR imaging platform, data band, and target type.

TABLE II
COMPARISON OF DIFFERENT METHODS ON ORI-RCNN AND FAST-RCNN (100% TRAINING DATA FINE-TUNING)

Method	Ori-RCNN		Fast-RCNN	
	AP (%)	F1 (%)	AP (%)	F1 (%)
Random-Baseline	79.77	73.10	74.16	70.26
ImageNet-Baseline	86.00	81.29	80.53	76.25
SimCLR	87.73	81.24	81.93	78.97
MoCo v2	85.06	79.77	77.43	74.07
SWAV	88.21	83.60	81.56	78.02
SimSiam	80.91	74.78	75.60	73.46
BYOL	87.89	83.47	81.24	80.87
RemoteClip	87.64	80.74	82.37	78.63
SwinUNet	80.19	76.48	82.32	78.86
SoftCon	81.56	73.29	81.89	78.79
MSFA	87.68	80.03	79.08	75.57
SARDet-CL (Ours)	**90.24**	**85.14**	**83.87**	**82.05**

TABLE III
SIMILARITY BETWEEN DIFFERENT DATASETS

Dataset	Platform	Band	Target Type	Resolution	**Similarity**
Self-Supervised	GF-3	C	airplane	1m	**Same**
SAR-Aircraft-1.0	GF-3	C	airplane	1m	**High**
ZKXT	GF-3	C	airplane	1m	**High**
MSAR-1.0	GF-3 HISEA-1	C	airplane, ship bridge, oiltank	1m	**Medium**
SADD	TerraSAR X		airplane	1m	**Low**

As shown in Table III, the SAR-AIRcraft-1.0, ZKXT dataset, and the self-supervised data are sourced from the same satellites, share the same data bands, and exclusively contain airplane targets. The only differences lie in the time and location of the shots, making them highly similar. In contrast, the MSAR-1.0 dataset is collected from two different satellites. A portion of this dataset, specifically the GF-3 data, exhibits characteristics that are highly similar to the self-supervised data. However, MSAR-1.0 also includes three additional target types—ship, bridge, and oil tank, which reduces its overall similarity compared to SAR-AIRcraft-1.0. The SADD dataset, by contrast, originates from the TerraSAR satellite, which uses a different band entirely. As a result, it is judged to have the lowest degree of similarity.

Similar to the settings in Table I in the previous section, to reduce the impact of supervised information on the final outcomes, only 20% of the data from each dataset was employed for fine-tuning, allowing for an assessment of the pretrained models' performance on nonhomologous datasets. The results are displayed in Table IV. The last row of the table demonstrates the difference in performance between the SARDet-CL method and the baseline method for each metric. The baseline model in this context refers to the ImageNet pretrained model for two main reasons: first, the baseline performance of random initialization is poor; second, the ImageNet pretrained model is widely regarded as the default configuration for improving model performance in computer vision tasks.

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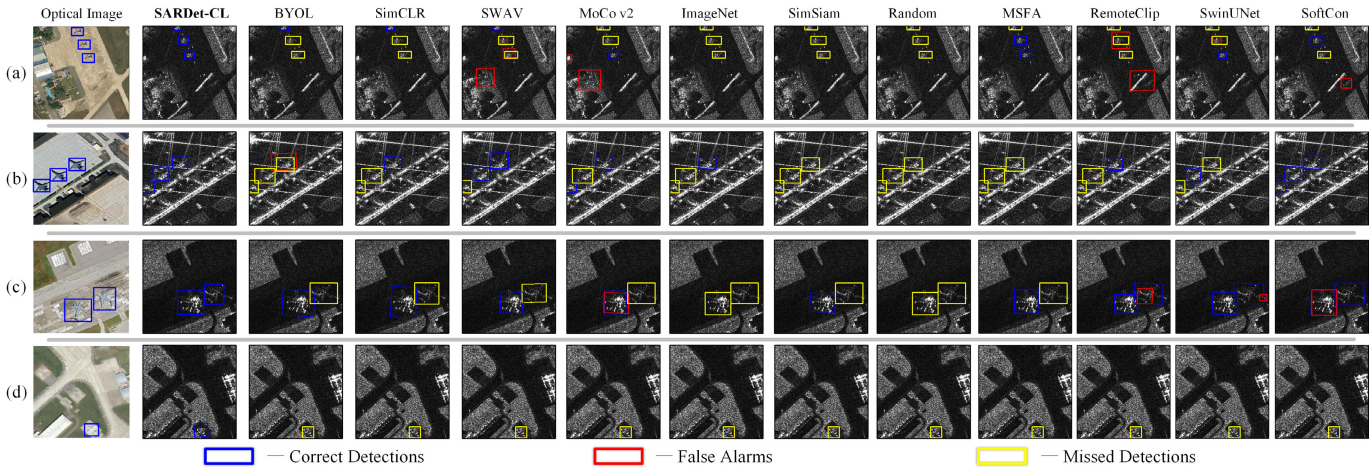


Fig. 9. Visualization of Ori-RCNN detection results: blue, yellow, and red represent correct detections, missed detections, and false positives, respectively.

TABLE IV
PERFORMANCE OF DIFFERENT METHODS ON NONHOMOLOGOUS DATASETS

Method	SAR-aircraft-1.0				MSAR-1.0				SADD				ZKXT			
	Ori-RCNN		Fast-RCNN		Ori-RCNN		Fast-RCNN		Ori-RCNN		Fast-RCNN		Ori-RCNN		Fast-RCNN	
	AP(%)	F1(%)	AP(%)	F1(%)	AP(%)	F1(%)	AP(%)	F1(%)	AP(%)	F1(%)	AP(%)	F1(%)	AP(%)	F1(%)	AP(%)	F1(%)
Random-Baseline	54.86	51.93	64.68	60.81	66.33	66.96	65.62	66.30	75.91	68.19	85.15	75.55	41.97	41.51	40.06	42.20
ImageNet-Baseline	80.16	74.70	74.88	69.24	68.93	70.55	66.32	68.32	85.95	79.77	85.74	81.11	69.70	64.59	65.64	60.67
SimCLR	82.58	75.47	80.19	76.56	68.61	70.57	69.52	71.16	83.10	78.55	85.76	81.18	74.58	67.74	73.81	69.28
MoCo v2	70.84	64.50	76.62	71.62	62.43	62.79	66.58	67.81	85.14	80.38	85.75	82.29	55.53	53.83	54.47	52.25
SWAV	81.70	77.35	80.54	76.00	71.33	72.66	69.83	71.13	85.82	79.06	85.80	79.91	75.71	70.79	70.97	66.04
SimSiam	65.45	63.38	61.40	58.84	65.77	65.53	66.39	62.75	82.52	75.59	85.32	77.31	53.74	53.50	41.92	42.38
BYOL	82.37	78.47	81.36	78.25	67.61	69.13	68.30	69.58	84.76	78.33	85.87	80.22	79.46	74.76	77.91	73.39
RemoteClip	48.38	49.36	53.17	53.59	48.35	49.69	50.21	50.57	76.38	64.16	81.19	73.20	35.96	31.27	19.94	12.99
SwinUNet	59.03	56.55	63.76	62.04	50.49	51.87	50.99	52.32	79.63	73.68	81.60	76.54	54.42	52.89	44.95	46.39
SoftCon	61.57	59.39	65.17	61.68	52.79	53.27	52.85	53.78	75.11	61.77	83.98	70.76	54.71	54.98	53.87	50.00
MSFA	79.11	73.72	78.82	75.61	52.86	54.16	56.58	57.44	85.81	80.87	86.31	81.17	74.16	68.64	65.91	62.81
SARDet-CL(Ours)	**86.06**	**84.53**	**86.79**	**86.21**	**72.42**	**75.63**	**73.79**	**75.23**	**87.97**	**83.41**	**87.68**	**84.39**	**81.97**	**78.37**	**79.03**	**73.89**
Diff with Baseline	**+5.90**	**+9.83**	**+11.91**	**+16.97**	**+3.49**	**+5.08**	**+7.47**	**+6.91**	**+2.02**	**+3.64**	**+1.94**	**+3.28**	**+12.27**	**+13.78**	**+13.39**	**+13.22**

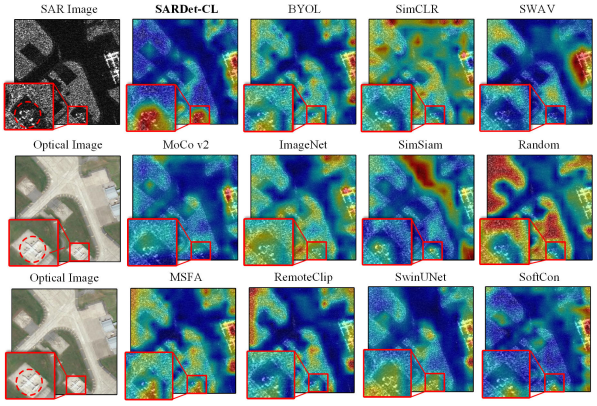


Fig. 10. Feature heatmaps generated by different methods, with red signifying higher attention and blue representing lower attention.

From the results, the SARDet-CL method outperforms other methods across the four nonhomologous datasets. Additionally, the effectiveness of various pretrained models on nonhomologous data shows considerable variation. For example, the SimCLR, SWAV, and BYOL methods surpass the baseline on the SAR-aircraft-1.0 dataset but perform poorly on the SADD dataset. Among the four datasets, the suboptimal methods are SimCLR, SWAV, MSFA, and BYOL, highlighting their inconsistent performance. In contrast, SARDet-CL demonstrates superior generalization, as its self-supervised signal design accounts for the unique characteristics of SAR images.

By examining the improvement of each indicator over the baseline method, SARDet-CL achieves better performance when the downstream task dataset is more similar to the self-supervised dataset. For example, on the SAR-aircraft-1.0 and ZKXT datasets, which have high data similarity, the overall improvement across all indicators is around 10%. In contrast, on the SADD dataset, which has low similarity, the improvement is only about 3%. Fortunately, SSL does not require labeled data, and it is easy to obtain a large amount of SAR data similar to the downstream task dataset.

Some detection results were presented to analyze the advantages of SARDet-CL. In Fig. 11(a), (b), (d), (e), (g), and (h), except for SARDet-CL, all other methods missed edge targets to varying degrees, among which the ImageNet pretrained model had the most serious problem. In Fig. 11(e) and (g), the count of missed targets is double that of other methods. This is likely due to the ImageNet dataset's bias

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TABLE V
PERFORMANCE ON LARGE IMAGES

Method	AP(%)	F1(%)	Method	AP(%)	F1(%)
Random-Baseline	77.68	68.25	BYOL	89.02	79.10
ImageNet-Baseline	84.53	76.40	RemoteClip	87.52	70.19
SimCLR	88.59	75.90	SwinUNet	78.63	71.83
MoCo v2	83.82	74.01	SoftCon	78.48	67.93
SWAV	88.22	78.72	MSFA	85.54	77.17
SimSiam	78.26	68.04	**SARDet-CL (Ours)**	**90.66**	**80.82**

toward central subjects, causing the model to overlook features at the periphery. As shown in Fig. 11(b) and (c), other methods struggle with the detection of similar targets, which is comparable to the issue seen in Fig. 9(a). Due to SAR's unique imaging mechanism, these targets exhibit distinct characteristics, and SARDet-CL demonstrates superior generalization. In Fig. 11(i), the scattered imaging process introduces noise interference in various parts of the aircraft, affecting correct detection and recognition. BYOL, SimCL, SimSiam, Random, RemoteClip, and GeRSP failed to detect the target, while SWAV and MOCO v2 detected the target but misidentified its type. The ImageNet-based method detected the target correctly but also produced a false alarm, incorrectly identifying scattering points on the wing as part of the aircraft. Similarly, the SWAV method misidentified the tail as an aircraft.

Additionally, Fig. 11(c), (d), (e), and (f) shows the detection of ship, oil tank, and bridge targets. SARDet-CL also achieves superior performance in detecting nonaircraft targets. This advantage stems from the ability of the self-supervised algorithm to learn intrinsic characteristics of SAR data, capturing both target and background information to extract distinguishable features. In contrast, supervised algorithms rely primarily on limited target information defined by human annotations. Fig. 11(j) and (k) shows the detection situation under a complex background, where the target is interfered with by strong sidelobe noise caused by buildings. Except for the SARDet-CL method, other methods have seriously missed detections and false alarms, among which MSFA performs relatively well. Consequently, the SARDet-CL method proves to be more effective for target detection tasks in heterogeneous SAR datasets.

3) Performance on Large Images: In this study, 16 large-scale GF-3 satellite SAR images with complex scenes were used to verify the accuracy and processing efficiency of the SARDet-CL method in practical applications. Since the pretraining did not change the structure of the downstream detector, the inference time of each pretraining method in the downstream task was almost the same. Table V quantitatively compares the performance of different methods, and the results show that the SARDet-CL method outperforms other methods in overall performance. In addition, Fig. 12 also shows some detection results for qualitative comparison and compares them with the detection results of the suboptimal method. In Chenggu Airport, SARDet-CL successfully detected small targets on the edge. In the second airport, SARDet-CL detected and identified correctly in a complex scene, while the suboptimal method had missed detections.

TABLE VI
RESULTS OF ABLATION STUDIES FOR EACH MODULE

Adv-Mask	Dync-Quantization	Multi-Temporal	3D-Structure	AP(%)	F1(%)
✓	✓	✓	✓	**70.29**	**67.17**
✗	✓	✓	✓	66.88	63.19
✓	✗	✓	✓	66.72	63.98
✓	✓	✗	✓	69.02	65.93
✓	✓	✓	✗	67.11	64.60
✓	✗	✗	✗	64.87	61.62
✗	✓	✗	✗	65.64	62.24
✗	✗	✓	✗	63.93	60.19
✗	✗	✗	✓	63.79	61.3
✗	✗	✗	✗	61.45	58.14

4) Module Ablation Study: Ablation experiments were used to evaluate the contribution of each module in the proposed SARDet-CL method. In Section III-B, the mask learning strategy based on the adversarial training (Adv-Mask) module and the Dync-Quantization module were designed to construct the self-supervised view. In Section III-C, the Multitemporal information learning and 3-D structural (3D-Structure) information compensation modules are used to provide feedback on the learning process of SARDet-CL. The starting points of the above modules in the design process do not contradict each other. The experimental setup is the same as in Section IV-C. The experimental data used are consistent with those used in Table I. Each module in SARDet-CL is ablated individually to assess its effectiveness. Table VI displays the results of the ablation experiments for each module.

From the experimental results, removing each of the four modules will reduce the performance of SARDet-CL, which demonstrates the effectiveness of each module. Among them, the Adv-Mask module and Dync-Quantization contribute more to the improvement of model performance, which can be improved by 4% points. These two modules can act on all samples during the working process. The 3-D-Structure module also significantly improves SARDet-CL. It acts on the negative samples in each batch and corrects its contrast loss with structural information. The introduction of the Multitemporal module can improve it by about 1% point, which means that multitemporal information is beneficial to the SSL task of SAR images. According to the judgment rules in Section III-C, the number of effective multitemporal alignment samples only accounts for 23% of the total self-supervised data. Due to the limitation of the amount of data on which the module works, the improvement is lower than that of the other three modules.

To further quantify the independent contributions of each module, we designed an ablation study using single modules to assess their impact on model performance. When all modules are turned off, the model is trained only by basic strategies such as rotation and folding in the classic CL method, ensuring the progress of SSL tasks. The experimental results indicate that each module positively contributes to model performance, with the Dync-Quantization strategy yielding the most significant improvement.

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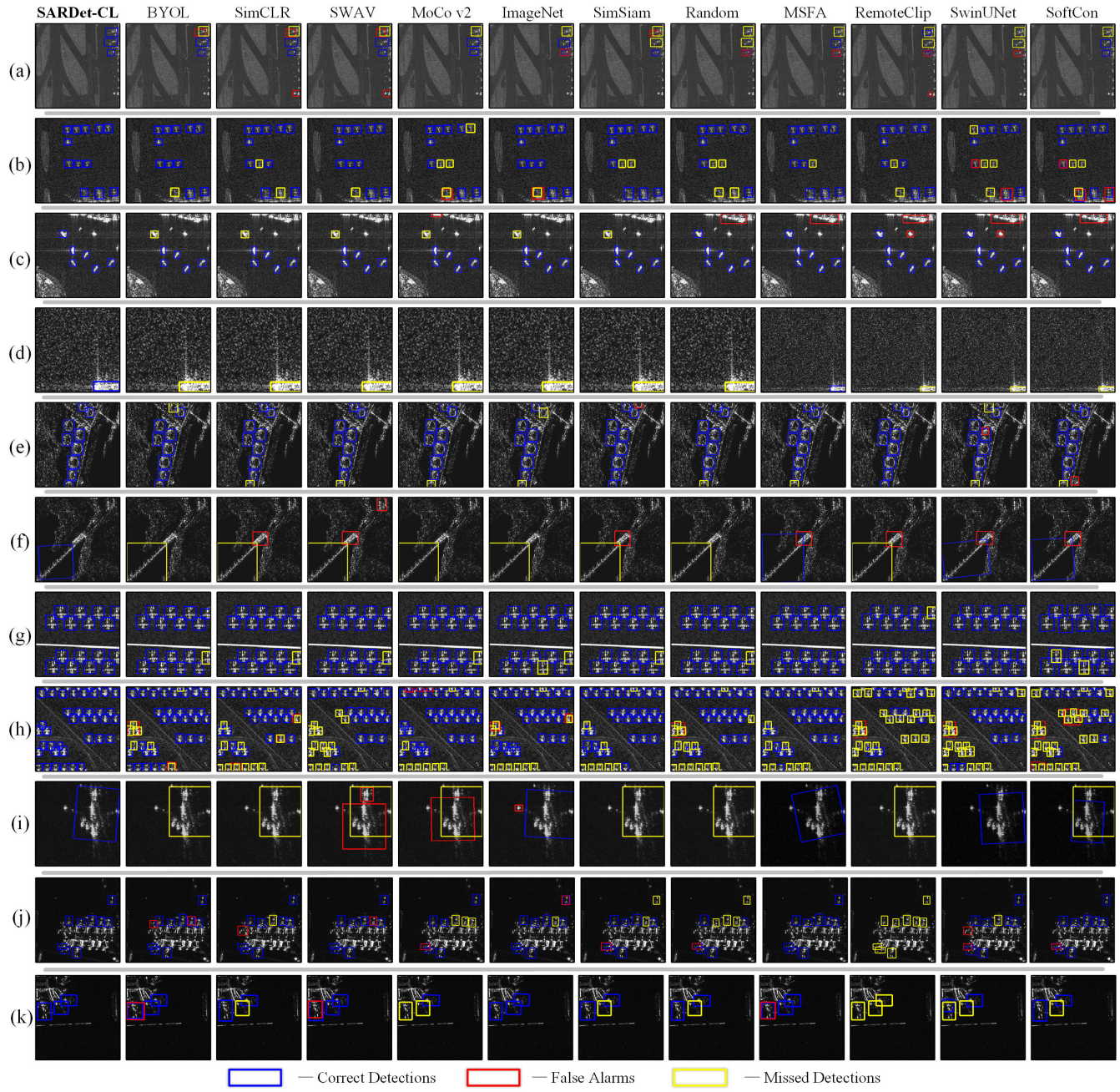


Fig. 11. Visualization of detection results of different methods on nonhomologous datasets. (a) and (b) SAR-Aircraft-1.0 dataset. (c)–(f) MSAR-1.0 dataset. (g)–(i) SADD dataset. (j) and (k) ZKXT dataset.

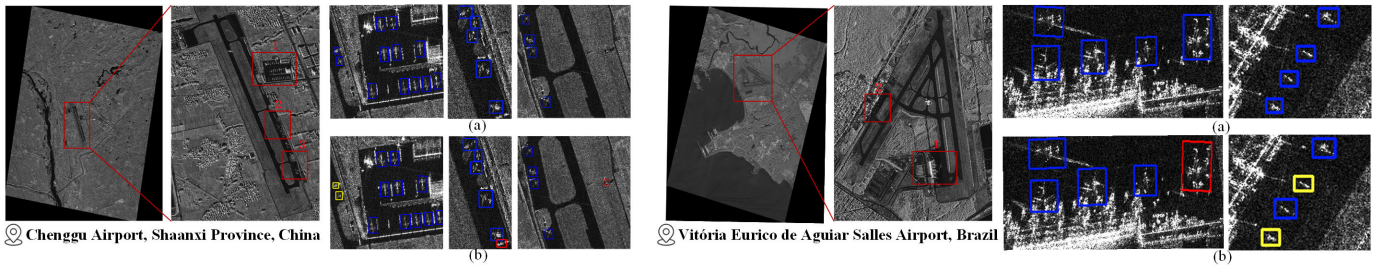


Fig. 12. Performance on large images. (a) Results of SARDet-CL. (b) Results of BYOL.

V. Conclusion

In this study, a self-supervised CL (SARDet-CL) framework for SAR aircraft target detection tasks is proposed. By introducing innovative strategies such as adversarial masking, Dync-Quantization, multitemporal information, and 3D-structural information compensation, the proposed method

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significantly enhances the performance of SSL in SAR aircraft target detection tasks.

The adversarial masking strategy enriches the feature learning process of the model by masking part of the image content, enabling it to progressively learn and generalize from incomplete target views. The Dync-Quantization mechanism based on spatial clustering enhances the model's adaptability to the unstable feature of targets, thereby improving the stability of target detection. In addition, spatiotemporally aligned samples are redefined as positive samples and 3D-structural information is introduced. Our framework enriches the SSL signal, allowing the model to effectively capture the spatiotemporal continuity and structural features present in SAR data, thereby more finely understanding and characterizing complex scenes.

Experimental results on multiple nonhomologous SAR datasets verify the advanced nature of the proposed SARDet-CL method. This study not only promotes the development of SAR image analysis but also provides new ideas for further exploration of SSL in remote sensing applications in the future.

REFERENCES

[1] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.

[2] X. Chen et al., "Context autoencoder for self-supervised representation learning," *Int. J. Comput. Vis.*, vol. 132, no. 1, pp. 208–223, Jan. 2024.

[3] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[4] F. Dellinger, J. Delon, Y. Gousseau, J. Michel, and F. Tupin, "SAR-SIFT: A SIFT-like algorithm for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 453–466, Jan. 2014.

[5] F. Xu and Y.-Q. Jin, "Imaging simulation of polarimetric SAR for a comprehensive terrain scene using the mapping and projection algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3219–3234, Nov. 2006.

[6] X. Xiao et al., "Aircraft detection and classification based on joint probability detector integrated with scattering attention," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 2, pp. 1722–1739, Apr. 2024.

[7] H. Jia, Q. Guo, J. Chen, F. Wang, H. Wang, and F. Xu, "Adaptive component discrimination network for airplane detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7699–7713, 2021.

[8] Y. Suo, Y. Wu, T. Miao, W. Diao, X. Sun, and K. Fu, "Adaptive SAR image enhancement for aircraft detection via speckle suppression and channel combination," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5219415.

[9] X. Sun, X. Li, D. Xiang, and C. Hu, "SAR vehicle image generation with integrated deep imaging geometric information," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104028.

[10] D. Xiang, W. Li, X. Sun, H. Wang, and Y. Su, "Sidelobe suppression for high-resolution SAR imagery based on spectral reshaping and feature statistical difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5211314.

[11] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[12] M. Caron, I. Misra, J. Mairal, P. Goyal, P. Bojanowski, and A. Joulin, "Unsupervised learning of visual features by contrasting cluster assignments," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9912–9924.

[13] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE Comput. Soc. Conf. Comput. Vision Pattern Recognit.*, Jun. 2021, pp. 15750–15758.

[14] J.-B. Grill et al., "Bootstrap your own latent-a new approach to self-supervised learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 21271–21284.

[15] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.

[16] C. Tao, J. Qi, G. Zhang, Q. Zhu, W. Lu, and H. Li, "TOV: The original vision model for optical remote sensing image understanding via self-supervised learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4916–4930, 2023.

[17] Z. Ren et al., "Self-supervised learning guided by SAR image factors for terrain classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406118.

[18] W. Li et al., "Predicting gradient is better: Exploring self-supervised learning for SAR ATR with a joint-embedding predictive architecture," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 326–338, Dec. 2024.

[19] Y. Wang, C. M. Albrecht, and X. X. Zhu, "Multilabel-guided soft contrastive learning for efficient Earth observation pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5644516.

[20] E. Schubert, J. Sander, M. Ester, H. P. Kriegel, and X. Xu, "DBSCAN revisited, revisited: Why and how you should (Still) use DBSCAN," *ACM Trans. Database Syst.*, vol. 42, no. 3, pp. 1–21, Sep. 2017.

[21] Z. Lei, F. Xu, J. Wei, F. Cai, F. Wang, and Y.-Q. Jin, "SAR-NeRF: Neural radiance fields for synthetic aperture radar multiview representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5221415.

[22] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, pp. 600–612, 2004.

[23] W. Zhirui, K. Yuzhuo, Z. Xuan, W. Yuelei, Z. Ting, and S. Xian, "SAR-AIRcraft-1.0: High-resolution SAR aircraft detection and recognition dataset," *J. Radars*, vol. 12 no. 4, pp. 906–922, 2023.

[24] J. Chen et al., "Large-scale multi-class SAR image target detection dataset-1.0," *J. Radars*, no. 1, 2022. [Online]. Available: https://radars.ac.cn/web/data/getData?dataType=MSAR

[25] P. Zhang et al., "SEFEPNet: Scale expansion and feature enhancement pyramid network for SAR aircraft detection with small sample dataset," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3365–3375, 2022.

[26] X. Zhao et al., "Tensor adaptive reconstruction cascaded with global and local feature fusion for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 607–620, 2025.

[27] X. Zhao, J. Huang, Y. Gao, and Q. Wang, "Hyperspectral target detection based on prior spectral perception and local graph fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13936–13948, 2024.

[28] X. Wang, A. Shrivastava, and A. Gupta, "A-fast-RCNN: Hard positive generation via adversary for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Sep. 2017, pp. 2606–2615.

[29] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[30] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.

[31] L. Scheibenreif, J. Hanna, M. Mommert, and D. Borth, "Self-supervised vision transformers for land-cover segmentation and classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1422–1431.

[32] Y. Li et al., "SARDet-100k: Towards open-source benchmark and toolkit for large-scale SAR object detection," in *Proc. 38th Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, pp. 128430–128461, 2024.



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